## Question

This problem involves acid-base equilibrium in precipitation-dissolution equilibrium.

To solve this problem, the following preliminary calculations are required:

First, calculate the $\mathrm{pH}$ of the solution obtained by dissolving $2.50\ \mathrm{g}\ \mathrm{KH_3(C_2O_4)_2{\cdot}2H_2O}$ in $100\ \mathrm{mL}$ of water.

Second, given that the $\mathrm{pH}$ of saturated $\mathrm{Ca(OH)_2}$ is $12.00$, calculate the $K_{\mathrm{sp}}$ of $\mathrm{Ca(OH)_2}$.

Based on these results, calculate the concentrations of each ion ($[\mathrm{H^+}]$, $[\mathrm{C_2O_4}^{2-}]$, $[\mathrm{Ca^{2+}}]$) in the solution obtained by mixing the solution described in the first step with saturated calcium hydroxide in a $1:1$ ratio.

Select the answer closest to the calculated values.

Relevant data: $\mathrm{p}K_{\mathrm{a1}}(\mathrm{H_2C_2O_4}) = 1.22$, $\mathrm{p}K_{\mathrm{a2}}(\mathrm{H_2C_2O_4}) = 4.19$, $\mathrm{p}K_{\mathrm{sp}}(\mathrm{CaC_2O_4}) = 8.70$.

**A.** $[\mathrm{H^+}] = 0.52\ \mathrm{mol \cdot L^{-1}}$

$[\mathrm{C_2O_4}^{2-}] = 6.2 \times 10^{-7}\ \mathrm{mol \cdot L^{-1}}$

$[\mathrm{Ca^{2+}}] = 2.5 \times 10^{-3}\ \mathrm{mol \cdot L^{-1}}$

**B.** $[\mathrm{H^+}] = 0.186\ \mathrm{mol \cdot L^{-1}}$

$[\mathrm{C_2O_4}^{2-}] = 3.23 \times 10^{-5}\ \mathrm{mol \cdot L^{-1}}$

$[\mathrm{Ca^{2+}}] = 6.16 \times 10^{-5}\ \mathrm{mol \cdot L^{-1}}$

**C.** $[\mathrm{H^+}] = 0.0225\ \mathrm{mol \cdot L^{-1}}$

$[\mathrm{C_2O_4}^{2-}] = 2.04 \times 10^{-4}\ \mathrm{mol \cdot L^{-1}}$

$[\mathrm{Ca^{2+}}] = 9.73 \times 10^{-6}\ \mathrm{mol \cdot L^{-1}}$

**D.** $[\mathrm{H^+}] = 0.0216\ \mathrm{mol \cdot L^{-1}}$

$[C_2O_4{}^{2-}] = 2.14 \times 10^{-4}\ \mathrm{mol \cdot L^{-1}}$

$[Ca^{2+}] = 9.48 \times 10^{-6}\ \mathrm{mol \cdot L^{-1}}$

**E.** $[H^+] = 0.0933\ \mathrm{mol \cdot L^{-1}}$

$[C_2O_4{}^{2-}] = 2.53 \times 10^{-5}\ \mathrm{mol \cdot L^{-1}}$

$[Ca^{2+}] = 7.88 \times 10^{-5}\ \mathrm{mol \cdot L^{-1}}$

**F.** $[H^+] = 0.0162\ \mathrm{mol \cdot L^{-1}}$

$[C_2O_4{}^{2-}] = 1.33 \times 10^{-5}\ \mathrm{mol \cdot L^{-1}}$

$[Ca^{2+}] = 0.005\ \mathrm{mol \cdot L^{-1}}$

**G.** None of the other options are correct

## Answer

**Correct Answer: D**

## Detailed Explanation

First Question

$$n(\mathrm{KH_3(C_2O_4)_2 \cdot 2H_2O}) = m/M = 9.83 \times 10^{-3}\ \mathrm{mol}$$

Concentration $c = n/V = 9.83 \times 10^{-2}\ \mathrm{mol \cdot L^{-1}}$

**CHECKPOINT** 1 PTS

Concentration $c = n/V = 9.83 \times 10^{-2}\ \mathrm{mol \cdot L^{-1}}$

The charge balance equation is:

$$[\mathrm{K^+}] + [\mathrm{H^+}] = [\mathrm{OH^-}] + [\mathrm{HC_2O_4}^-] + 2[\mathrm{C_2O_4}^{2-}]$$

**CHECKPOINT** 2 PTS

The charge balance equation is: $[\mathrm{K^+}] + [\mathrm{H^+}] = [\mathrm{OH^-}] + [\mathrm{HC_2O_4}^-] + 2[\mathrm{C_2O_4}^{2-}]$

Let $[\mathrm{H^+}] = x\ \mathrm{mol/L}$

Based on the distribution fraction, we have:

$$[HC_2O_4^-] = c_0(H_2C_2O_4) \times (10^{-1.22}x)/(x^2 + 10^{-1.22}x + 10^{-1.22} \times 10^{-4.19})$$

$$[C_2O_4^{2-}] = c_0(H_2C_2O_4) \times (10^{-1.22} \times 10^{-4.19})/(x^2 + 10^{-1.22}x + 10^{-1.22} \times 10^{-4.19})$$

**CHECKPOINT** 3 PTS

Based on the distribution fraction, we have: $[HC_2O_4^-] = c_0(H_2C_2O_4) \times (K_{a1}[H^+])/([H^+]^2 + K_{a1}[H^+] + K_{a1} \times K_{a2})$ , $[C_2O_4^{2-}] = c_0(H_2C_2O_4) \times (K_{a1} \times K_{a2})/([H^+]^2 + K_{a1}[H^+] + K_{a1} \times K_{a2})$

Here, $c_0(H_2C_2O_4) = 2c$

Substituting into the above equations, we solve for $x = 0.0314 \text{ mol} \cdot L^{-1}$

Thus, $pH = 1.50$

Second Question

Calculate $K_{sp}$ using the pH of saturated calcium hydroxide:

$$K_{sp}(Ca(OH)_2) = [OH^-]^2 \times [Ca^{2+}] = 1/2[OH^-]^3 = 5 \times 10^{-7}$$

**CHECKPOINT** 1 PTS

$K_{sp}(Ca(OH)_2) = 5 \times 10^{-7}$

After mixing the two solutions described in the problem:

$$[K^+] = 1/2c = 4.91 \times 10^{-2} \text{ mol} \cdot L^{-1}$$

The concentration of saturated calcium hydroxide is $0.005 \mathrm{mol} \cdot \mathrm{L}^{-1}$, while the oxalate concentration in the first solution is $c = 3 \times 9.83 \times 10^{-2}\,\mathrm{mol} \cdot \mathrm{L}^{-1} = 0.295 \mathrm{mol} \cdot \mathrm{L}^{-1}$. Therefore, calcium hydroxide is insufficient after mixing.

**CHECKPOINT** 1 PTS

Calcium hydroxide is insufficient after mixing

Comparing the solubility product constant with the solution acidity, it can be concluded that, under insufficient calcium hydroxide conditions, calcium hydroxide completely reacts and converts to calcium oxalate, i.e., $c_0(\mathrm{H_2C_2O_4}) = 9.33 \times 10^{-2}\,\mathrm{mol} \cdot \mathrm{L}^{-1}$

**CHECKPOINT** 1 PTS

Calcium hydroxide completely reacts and converts to calcium oxalate

Since the total oxalate concentration is much higher than the total calcium concentration, the calcium ion concentration can be neglected based on the precipitation-dissolution equilibrium.

**CHECKPOINT** 1 PTS

Calcium ion concentration can be neglected

Thus, the charge balance equation is the same as in the first question:

$$[\mathrm{K^+}] + [\mathrm{H^+}] = [\mathrm{OH^-}] + [\mathrm{HC_2O_4}^-] + 2[\mathrm{C_2O_4}^{2-}]$$

Following the same logic as the first question and using the distribution fraction, we calculate:

$$[H^+] = 0.0216\ \text{mol} \cdot L^{-1}$$

**CHECKPOINT** 1 PTS

Following the same logic as the first question and using the distribution fraction, we calculate: $[H^+] = 0.0216\ \text{mol} \cdot L^{-1}$

This naturally yields:

$$[C_2O_4{}^{2-}] = 2.14 \times 10^{-4}\ \text{mol} \cdot L^{-1}$$

Based on the precipitation-dissolution equilibrium, it follows that:

$$[Ca^{2+}] = 9.48 \times 10^{-6}\ \text{mol} \cdot L^{-1}$$

**CHECKPOINT** 1 PTS

$[Ca^{2+}] = 9.48 \times 10^{-6}\ \text{mol} \cdot L^{-1}$

**CHECKPOINT** 1 PTS

$[C_2O_4{}^{2-}] = 2.14 \times 10^{-4}\ \text{mol} \cdot L^{-1}$

**CHECKPOINT** 1 PTS

$[H^+] = 0.0216\ \mathrm{mol \cdot L^{-1}}$

Thus, option D is correct.

**CHECKPOINT** 1 PTS

Option D is correct